

NRIPENDRO BISWAS

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PROFILE SUMMARY

Undergraduate Electrical and Electronic Engineering student at Shahjalal University of Science and Technology (SUST) with research experience in machine learning, time-series forecasting, and perovskite solar cell modeling. Author/co-author of multiple IEEE conference and Scopus-indexed publications. Skilled in Python, SCAPS-1D, and predictive modeling, with research interests in data science, AI-driven energy systems, semiconductor devices, and renewable energy technologies.

EDUCATION

Shahjalal University of Science and Technology (SUST)

B.Sc in Electrical and Electronic Engineering

Mar 2022 – present

Sylhet, Bangladesh

Murari Chand (MC) College

Higher Secondary Certificate (HSC); GPA: 5.00/5.00

2018 – 2020

Sylhet, Bangladesh

Jalalabad Cantonment Public School & College

Secondary School Certificate (SSC); GPA: 5.00/5.00 (with 92.61% marks)

2016 – 2018

Sylhet, Bangladesh

RESEARCH INTERESTS

Time-series Forecasting, Machine Learning for Energy Systems, Perovskite Solar Cells, Semiconductor Devices.

PUBLICATIONS

Journal Articles

❖ N. Biswas, M. A. Mahdi, M. D. Gupta, Electricity demand forecasting for tetouan, morocco using darts in machine-learning: A comparative study of forecasting models, Results in Engineering (2026) 110289.

Conference Proceedings

❖ N. Biswas, I. Hossain, M. F. Jaman, M. M. Hassan and M. A. Mahdi, "Electricity Demand Forecasting of Bangladesh Based on Darts Temporal Fusion Transformer (DTFT)," 2025 7th International Conference on Electrical Information and Communication Technology (EICT), Khulna, Bangladesh, 2025, pp. 1-6.

❖ M. A. Mahdi, N. Biswas and M. D. Gupta, "Comparison of Machine Learning Models for Accurate Prediction of Power Conversion Efficiency in Perovskite Solar Cell," 2025 7th International Conference on Electrical Information and Communication Technology (EICT), Khulna, Bangladesh, 2025, pp. 1-6.

❖ N. Biswas, A. Rijvee, M. A. Mahdi, A. Azam, M. T. Hossain, Bser-bengali speech emotion recognition using mfcc features based on subesco and banglaser: A comparative study of machine learning models, in: 8th International Conference on Engineering Research, Innovation, and Education 2025 (ICERIE 2025), Atlantis Press, 2025, pp. 636–644.

❖ N. Biswas, M. A. Mahdi and T. Islam, "BSER-Bengali Speech Emotion Recognition Based on SUBESCO Dataset," 2024 13th International Conference on Electrical and Computer Engineering (ICECE), Dhaka, Bangladesh, 2024, pp. 673-678.

❖ N. Biswas, M. A. Mahdi and T. Islam, "A Deep Learning Approach to Forecast Electricity Demand in Sylhet of Bangladesh," 2024 International Conference on Innovations in Science, Engineering and Technology (ICISSET), Chittagong, Bangladesh, 2024, pp. 1-6.

RESEARCH EXPERIENCE

❖ Undergraduate Student Researcher, Shahjalal University of Science and Technology (SUST), Sylhet

Supervisor: Mohona Das Gupta, Lecturer, Department of Electrical & Electronic Engineering, SUST

2024 – Present

❖ Developed electricity demand forecasting pipelines using Darts with TFT, TSMixer, TiDE, Random Forest, and FB Prophet models on ENTSO-E and Tetouan datasets. Performed pre-processing, feature engineering, hyperparameter tuning, and comparative evaluation using MAE, RMSE, and MAPE metrics. Conducting SCAPS-1D simulation and parameter optimization for lead-free perovskite solar cells.

❖ Undergraduate Student Researcher, Adnan Research Group (ARG)

Supervisor: Dr. Md. Mohsinur Rahman Adnan, Assistant Professor, Department of Electrical & Electronic Engineering, BUET

2024 – Present

❖ Conducting comparative research on transformer-based and deep learning forecasting models using the ENTSO-E dataset.

ACADEMIC PROJECTS

❖ ASIC Design Project | Design and Verification of 1-Bit and 2-Bit Magnitude Comparators Using Cadence Virtuoso.

❖ Perovskite Solar Cell Modeling | Numerical simulation and ML-based optimization using SCAPS-1D.

❖ BenSER | Real-Time Bengali Speech Emotion Recognition using ML.

INDUSTRIAL TRAINING

❖ Neural Semiconductor, Dhaka, Bangladesh

Industrial Attachment - VLSI & Semiconductor Design

19 Dec 2025 – 26 Dec 2025

❖ RTL design, logic synthesis, ASIC PnR flow, Fundamentals of analog and RF circuit design.

❖ Training Institute for Chemical Industries (TICI) , Ghorashal, Narsingdi, Bangladesh Industrial Training on Electrical Engineering and Instrumentation	6 Dec 2025 – 18 Dec 2025
❖ Electrical instrumentation, PLC, SCADA, DCS, protection systems, and industrial safety.	
❖ Hands-on experience with generators, motors, and transformers.	
❖ Consumers Association of Bangladesh (CAB) , Bangladesh Training on Solar System Planning, Design, Installation, and Operation	Sept 2024 – Oct 2024
❖ PV system design, simulation, installation, safety practices, and O&M strategies.	
❖ Hands-on laboratory training at DIU Renewable Energy Lab.	

ACHIEVEMENTS

❖ Finalist , BTCLF Youth Talent Hunt-2022
❖ 7th in National Round and 1st in Sylhet region.

ACADEMIC HONORS & SCHOLARSHIPS

❖ Higher Secondary Certificate (HSC) Board Scholarship — 2020
❖ Secondary School Certificate (SSC) Board Scholarship — 2018

LEADERSHIP EXPERIENCES

❖ Secretary IEEE SUST Student Branch , Sylhet, Bangladesh	April 2024 – Present
❖ IT Secretary Columbia University NEBDHub NSDC SUST Chapter , Sylhet, Bangladesh	Feb 2025 – Present
❖ Treasurer RoboSUST , Sylhet, Bangladesh	April 2023 – Present

VOLUNTEERING EXPERIENCE

❖ Ambassador IEEE Bangladesh Section Conferences 2025
❖ Mentor RoboSUST PyChamp 2024

TECHNICAL SKILLS

Programming	Python, C, VHDL, Arduino
Simulation & Design	Cadence, MATLAB, SCAPS-1D, CST, LTspice
Design Tools	KiCad, Proteus, AutoCAD
Data Tools	Excel, OriginLab

REFERENCES

Mohona Das Gupta Lecturer Department of Electrical and Electronic Engineering Shahjalal University of Science and Technology, Sylhet Contact: mohona-eee@sust.edu	Dr. Md. Mohsinur Rahman Adnan Assistant Professor Department of Electrical and Electronic Engineering Bangladesh University of Engineering and Technology (BUET), Dhaka Contact: mmradnan@eee.buet.ac.bd
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